C++

Lecture II

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Quick Recap

- Last time:
  - Materials setup
  - **WIP** TetrisBlock Actor: C++ and BP
    - Setting material color, hiding / showing
- **TODO**
  - Finish TetrisBlock Actor C++
  - TetrisPlayer
    - Utilize Enhanced input
    - Create callbacks for input keybinds and verify functionality
  - TetrisGameMode
    - Board visualization
    - Delegate binding with Pawn
    - Implement game logic

TetrisBlock - Functions

- We'd like to have a basic cube mesh on the actor, assign the dynamic material to the mesh and be able to read and assign a mesh color
- Constructor
  - Load a default cube mesh from the engine
- `BeginPlay()`
  - Check if material is assigned
  - Create dynamic material instance and set the material to the mesh component
  - (Later) Set visibility of the actor to `false`
- Create Getter & Setter for `BlockColor` variable
  - Setter will also set material color - `SetVectorParameterValue()`
  - If the new color is **Black**, hide the actor
- Test the behavior by assigning a pseudorandom color to the block

Asserts in UE

- `check() / checkf()`
  - Behaves like assert in C/C++
  - Failed condition **stops execution**, shows dialog, breakpoint
  - Does not evaluate the condition in optimized build
- `verify() / verifyf()`
  - Similar to check, **stops execution**
  - Makes sure that evaluated condition is not optimized out in optimized build configurations
- `ensure() / ensureMsgf()`
  - Similar to verify, but **does not stop execution** -> non fatal errors
  - Failure reported only once
    - Use `ensureAlways()` if you want to report always

Next steps

- We have the board building block set up - player input and the game logic left
- Game Mode
  - Spawn the board
  - (Later) Put all of our game logic in there
  - (Later) Player's Pawn has access to the game mode:
    - Direct access - quick implementation, but messy solution
    - Delegates - slower implementation, but preferred way
- Player input
  - Utilize Enhanced Input
  - Setup input actions
  - Create `TetrisGameMode` and set up our custom Pawn
  - Print out debug messages into the Output Log to verify functionality
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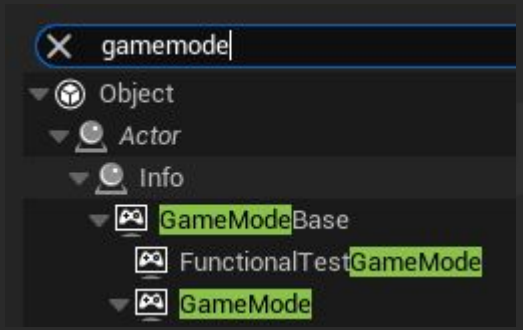
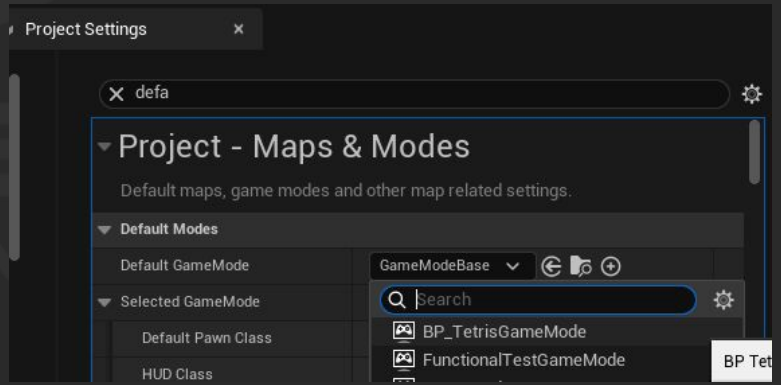
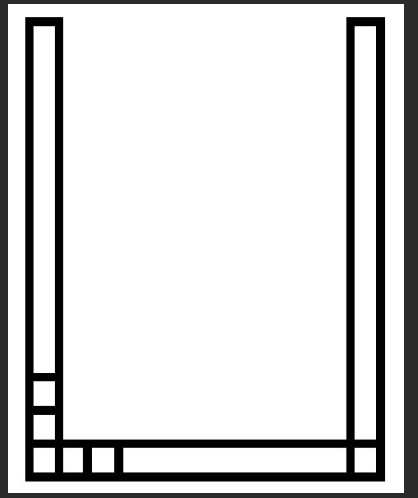
Editor - Create and Set GameMode

- Create new C++ class:
  - `TetrisGameMode` (**`GameModeBase`**)
    - Game mode is a level persistent class accessible from everywhere
    - For level transfer persistent class, use GameInstance
- Create `BP_TetrisGameMode` based on the newly created C++ class
- Make the `BP_TetrisGameMode` the default game mode in Project Settings





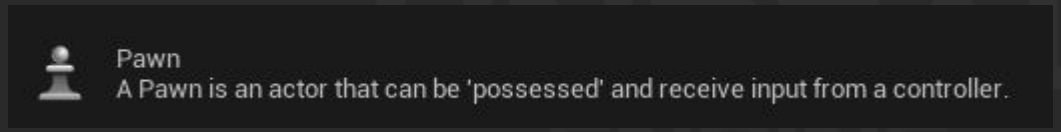
TetrisGameMode



- Variables:
  - `TSubclassOf<ATetrisBlock> BlockClassToSpawn;`
  - `TArray<ATetrisBlock*> Board;`
- `Functions:`
  - `BeginPlay()`
    - Check if the variable `BlockClassToSpawn` is assigned
    - Spawn **16x20** (WxH) grid of `BP_TetrisBlock` actors
      - Block size = 100 `Unreal Units` (distance between them)
      - Store spawned actors in the `Board` array
    - After you verify blocks are spawned:
      - Set color of blocks to **White** unless the block is on a border, except the top row non-border blocks - then color is **Black.** When color is **Black**, hide the actor.

Editor - Create and Set TetrisPawn

- Create new C++ class:
  - `TetrisPlayer` (Pawn)
- Create `BP_TetrisPlayer` based on the newly created C++ class
  - Set it as Default Pawn in the `BP_TetrisGameMode`
- When you hit Play, you won't be able to move around anymore
  - Move `PlayerStart` and rotate it in a way that will fit our scene



Editor - Enhanced Input

- Create & Navigate to a folder ‘Input’ in the Content Browser
- Create mapping context: `IMC_TetrisMappingContext`
- Create 3 Input Actions:
  - `IA_MoveBlock`
    - Axis1D (float)
    - Triggers: Pressed
  - `IA_PlaceBlock`
    - Digital (bool)
    - Triggers: Pressed + Released
  - `IA_RotateBlock`
    - Digital (bool)
    - Triggers: Pressed
- Map them to keys of your choice in the `TetrisMappingContext`
  - A key: `Negate` modifier to move block left

Player’s Pawn - Variables

```cpp
// This Include
#include "TetrisPlayer.h"

// Engine Include
#include "EnhancedInputComponent.h"
#include "EnhancedInputSubsystems.h"
#include "InputActionValue.h"
```

- Connection to the Enhanced Input
  - Add includes in `TetrisPlayer.cpp`:
- Add private member variables to `TetrisPlayer.h`
  - `UInputMappingContext* TetrisMappingContext;`
  - `UInputAction* MoveBlock;`
  - `UInputAction* PlaceBlock;`
  - `UInputAction* RotateBlock;`
- Forward declare used types (all are classes)
- `UPROPERTY(EditAnywhere, Category = Input)`
  - All UPROPERTY specifiers explained

Learning material

- UE Naming convention
- Programming in UE C++, the whole documentation
- Balancing BPs and C++
- Getting started with Unreal Engine
- Asserts in UE
- Objects in UE